Practical - 13

PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload

Problem Definition: Create simple PHP JSON API endpoints for events or students and consume them using Fetch API. Perform list, add, update, and delete operations asynchronously without full page reload.

A. Theory

1. What is a JSON API?

Answer: A JSON API allows a frontend and backend to exchange structured data in JSON format.

2. What is Fetch API?

Answer: Fetch is a JavaScript interface for making asynchronous HTTP requests and handling responses using promises.

3. What does CRUD without page reload mean?

Answer: JavaScript sends a request to PHP, receives JSON, and updates only the relevant part of the page instead of reloading the whole document.

4. Why use HTTP status codes and error handling?

Answer: Status codes communicate outcomes consistently, while structured error handling helps the frontend display useful messages and diagnose failures.

B. Implementation / Procedure

1. API endpoint

Create a PHP endpoint that sets Content-Type: application/json, checks the HTTP method/action, and returns JSON for every outcome.

2. GET list

Connect to MySQL using MySQLi and return a list of students or events as JSON.

3. Create/update/delete

Read JSON request bodies using json_decode(file_get_contents('php://input'), true) when appropriate. Validate values and use prepared INSERT, UPDATE, and DELETE statements.

4. Frontend fetch

Use fetch() with the appropriate method, headers, and JSON.stringify() body. Check response.ok and parse response.json().

5. Update interface

Render returned records into the page and show loading, success, empty, and error states. Avoid inserting untrusted data with unsafe innerHTML.

6. Test API

Use browser DevTools Network tab to verify request method, payload, status code, response JSON, and that the page did not reload.

C. Test Cases

Test Case	Input / Action	Expected Result
List records	Open page	GET returns valid JSON and list renders
Add record	Submit valid data	POST succeeds and list updates without reload
Invalid data	Submit missing/invalid fields	API returns an error and no row is inserted
Update record	Edit a record	Database and UI show updated data
Delete record	Delete a record	Record disappears without full-page reload

D. Key Questions / Analysis / Interpretation

1. Do PHP endpoints return valid JSON responses?

Answer: The endpoint sets an application/json content type and returns a consistent JSON object, including success status, message, and data where applicable.

2. Are GET and POST requests handled properly?

Answer: GET is used to retrieve data; POST can create records. Update and delete may use POST with an explicit action or appropriate HTTP methods, with server-side validation.

3. Are loading, success, and error states shown on the frontend?

Answer: The interface shows loading feedback during a request, confirms successful operations, and displays an error if the request or server fails.

4. Is input validation performed before database operations?

Answer: The PHP endpoint validates every field and identifier before running a prepared database query. Frontend validation is not sufficient on its own.

M. Evidence of Work (Screenshots)

25DCE050

StudentHub Portal

Home | About | AJAX CRUD | Login | Dashboard | Events | Profile | Contact | FAQ | Feedback

Light / Dark

Practical 13: Asynchronous Fetch-Based CRUD

Consuming PHP JSON API without Page Reload

Student record created successfully.

Add New Student via API

Full Name

Email Address

Mobile Number

Course

B.Tech Computer Engineering

Year

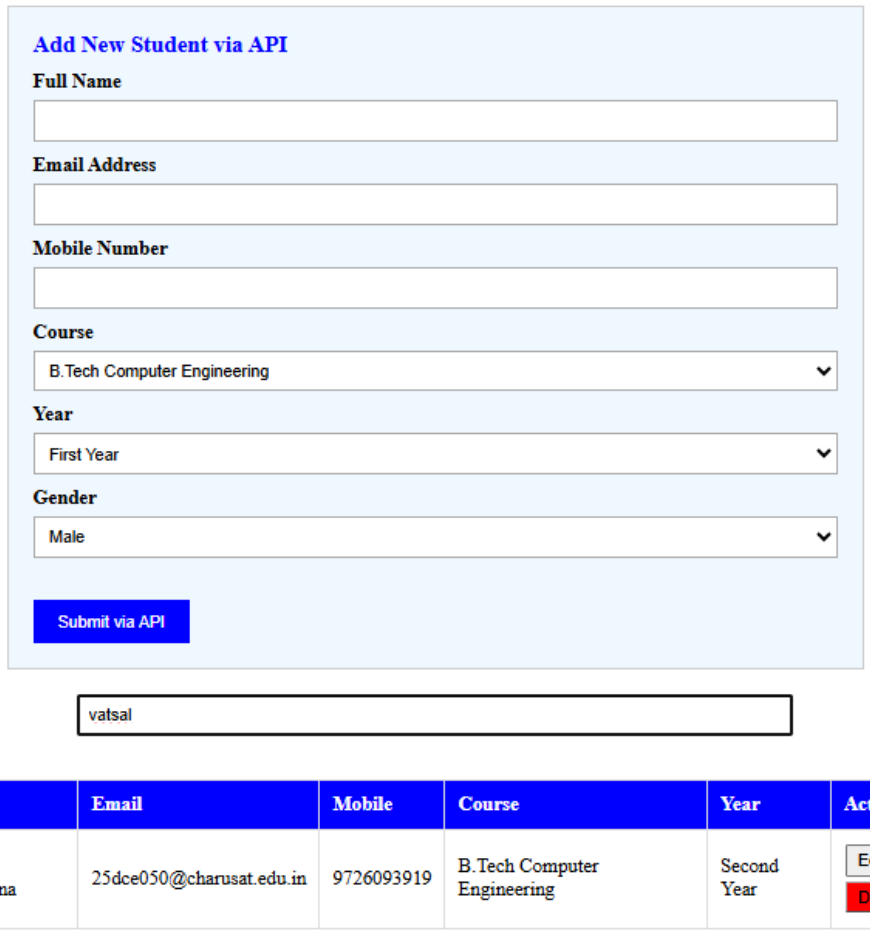
First Year

Gender

Male

Submit via API


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Practical 13: Asynchronous Fetch-Based CRUD
Consuming PHP JSON API without Page Reload
Add New Student via API
Full Name
Email Address
Mobile Number
Course
B.Tech Computer Engineering
Year
First Year
Gender
Male
Submit via API
Debounced live search...
ID
Name
Email
Mobile
Course
Year
Actions
22
archi bhatt
25dce009@charusat.edu.in
9316806552
B.Tech Computer Engineering
Second Year
Edit
Delete
21
vatsal makwana
25dce050@charusat.edu.in
9726093919
B.Tech Computer Engineering
Second Year
Edit
Delete
20
Kavya Barot
kavya.barot@charusat.edu.in
9825010203
B.Tech Information Technology
First Year
Edit
Delete
19
Pranav Makwana
pranav.makwana@charusat.edu.in
9825001234
B.Tech Mechanical Engineering
Second Year
Edit
Delete
18
Meera Solanki
meera.solanki@charusat.edu.in
9825090123
B.Tech Civil Engineering
First Year
Edit
Delete



ID	Name	Email	Mobile	Course	Year	Actions
21	vatsal makwana	25dce050@charusat.edu.in	9726093919	B.Tech Computer Engineering	Second Year	Edit Delete

N. GitHub Repository Link

https://github.com/vatsal-1950/StudentHUB.git